FACTORYMIND CDT

PROJECT DOCUMENTATION — INITIAL SUBMISSION

An Uncertainty-Aware Cognitive Digital Twin for Evidence-Grounded Manufacturing Diagnosis, Simulation, and Decision Support

Project Name	FactoryMind CDT
Track	Machine Learning Engineer
Team Leader	محمد عادل رمضان عبد الحميد
Team Size	6 students
Required Sections	Project Planning & Management; Literature Review; Requirements Gathering
Initial Documentation Deadline	Guidelines list 16 Oct 2026; initiative announcement states 19 Oct 2026
Final Project Deadline	6 Dec 2026 as stated by team; verify with instructor/LMS
Status	Proposed design; no implementation or real-factory validation claimed

Important: Proposed roles, dates, KPIs, and technical choices must be confirmed by the team and instructor.

Team Roster and Responsibility Allocation

The names below follow the Arabic team roster provided. Role allocation is the team's proposed workstream split and should be confirmed by all members.

Student ID	Official Name (Arabic)	Assigned Workstream
21003072	محمد عادل رمضان عبد الحميد	Team Lead; System Architecture; API Contracts & Integration; Integration QA
21140977	محمد أسامة عبد المحسن عبد العزيز	Data Engineering; Dataset Audit; Data Dictionary; Validation & Preprocessing
21046871	بيشوي سعد بشاره مخائيل	Time-Series ML; Anomaly Detection; Forecasting; Model Evaluation
21010230	أحمد عصام راشد عبد العال	NLP; Technical Documentation; Retrieval & Reranking
21029403	علي معتصم ممدوح عبد	Evidence Graph; Root Cause Analysis (RCA); Uncertainty & Abstention
21128637	زياد أحمد ربيع علي	Bounded Digital Twin; Impact Calculator; Demo UI

1. Project Planning & Management

1.1 Project Proposal and Overview

Problem: Manufacturing teams may investigate equipment anomalies using fragmented sensor readings, maintenance records, quality reports, and operating context. An isolated alert may lack traceable evidence.

Proposed solution: A decision-support prototype combining authorized manufacturing data, ML anomaly analysis, evidence retrieval, explicit uncertainty, and one bounded scenario simulation. It provides candidate explanations for human review and does not control machinery.

Target users: Maintenance engineers, production engineers, quality engineers, and supervisors.

1.2 Objectives

- Prepare a reproducible pipeline for one selected dataset/simulator.
- Build and evaluate a baseline anomaly-detection or failure-classification method, depending on labels.
- Link incident results to evidence and label candidate causes as hypotheses.
- Demonstrate one bounded what-if scenario with stated assumptions.
- Measure model quality, false alarms, evidence traceability, and response time.

1.3 Scope and Boundaries

Included: one use case/dataset, data cleaning, baseline ML, incident-review interface, evidence links, one bounded scenario, evaluation, documentation. Excluded: direct machine/PLC control, automatic setpoint changes, unsupported proprietary integrations, and untested real-factory claims.

1.4 Proposed Work Plan and Milestones

Proposed internal plan based on team-stated 6 December 2026 final deadline. The guideline lists 16 October for initial DEPI sections; the announcement supplied by the team says 19 October. Confirm applicable deadline with instructor.

Period	Milestone	Deliverable
19–25 Oct	Scope and data	Use case, dataset, data dictionary, backlog, acceptance criteria
26 Oct–1 Nov	Data and baseline	Preprocessing, split strategy, baseline metric, limitations
2–8 Nov	ML prototype	Model, evaluation, error review
9–15 Nov	Evidence investigation	Incident view, source records, uncertainty
16–22 Nov	Simulation and UI	Bounded scenario and dashboard
23–29 Nov	Integration and tests	Integrated prototype and test results

30 Nov–4 Dec	Final evaluation	Results, limitations, guide, report, slides
5–6 Dec	Final handoff	Repository, README, demo, submission

1.5 Task Assignment & Roles

Task allocation follows the team's proposed six-workstream division. The team should approve these assignments and designate a backup reviewer for each workstream.

Team Member	Assigned Responsibility
محمد عادل رمضان عبد الحميد	Team Lead; System Architecture; API Contracts & Integration; Integration QA
محمد أسامة عبد المحسن عبد العزيز	Data Engineering; Dataset Audit; Data Dictionary; Validation & Preprocessing
بيشوي سعد بشاره مخائيل	Time-Series ML; Anomaly Detection; Forecasting; Model Evaluation
أحمد عصام راشد عبد العال	NLP; Technical Documentation; Retrieval & Reranking
علي معتصم ممدوح عبد	Evidence Graph; Root Cause Analysis (RCA); Uncertainty & Abstention
زياد أحمد ربيع علي	Bounded Digital Twin; Impact Calculator; Demo UI

1.6 Resources and Dependencies

- Python analysis/ML tools and GitHub; choose dashboard framework after scope/data confirmation.
- Public or authorized dataset/simulator; check license, provenance, labels, units, missing values.
- Six team members, instructor, and domain reviewer if available.
- Dependencies: data access, stable environment, agreed metrics, feedback.

1.7 Risk Assessment & Mitigation

Risk	Impact	Mitigation
No labels	High	Choose early; use anomaly detection if needed
Data leakage	High	Appropriate split; fit preprocessing on training only
False alarms	High	Report metrics and errors; human review
Broad scope	High	One bounded scenario; state assumptions
Integration delay	Medium	Define interfaces; integrate weekly
Privacy/licensing	High	Authorized data; record license
Schedule slippage	High	MVP, milestones, escalate blockers

1.8 KPIs

KPI	Measurement	Provisional acceptance
Model quality	Task-appropriate metrics	Set after data/baseline inspection
False alarms	False positives/reviewed sample	Report rate and examples
Evidence traceability	Evidence items with source IDs	100% source identifiers
Response time	Incident selection to results	Set after profiling
Reproducibility	Second member runs README	One clean successful run
Testing	Critical tests passed	All critical workflow tests pass

1.9 Workstream Dependencies and Integration

The six workstreams will be integrated through agreed data schemas and API contracts. Work can proceed in parallel once the shared contracts are agreed; the sequence below describes the main dependencies.

- Data Engineering (محمد أسامة عبد المحسن عبد العزيز) defines the dataset schema, data dictionary, validation checks, and reproducible prepared-data output.
- Time-Series ML (بيشوي سعد بشاره مخائيل) consumes validated data and returns anomaly/forecast outputs with timestamps, model version, and evaluation results.

- NLP and Retrieval (أحمد عصام راشد عبد العال) prepares technical-document chunks and returns ranked evidence with source identifiers and retrieval scores.
- Evidence Graph and RCA (علي معتصم ممدوح عبد) combines model outputs and retrieved evidence into candidate cause hypotheses, records supporting or conflicting evidence, and represents uncertainty or abstention.
- Digital Twin and UI (زياد أحمد ربيع علي) consumes agreed outputs and runs bounded what-if scenarios; the interface displays assumptions and does not control real equipment.
- Team Lead (محمد عادل رمضان عبد الحميد) defines and reviews shared schemas/API contracts, integrates components, and coordinates integration tests.

Shared contract fields to agree before implementation: event_id, timestamp, asset_id (if available), data_source_id, model_version, anomaly_score or predicted_label (when supported), evidence_source_id, candidate_hypothesis, uncertainty_status, scenario_id, and evaluation metadata. If a field is unavailable in the selected dataset, mark it unavailable rather than inventing a value.

2. Literature Review

2.1 Review Focus and Method

Focused initial review of manufacturing digital-twin frameworks, predictive maintenance, ML diagnosis, and industrial data integration; not exhaustive.

2.2 Similar Work and Comparison

Source	Contribution	Limitations / relevance
ISO 23247-1:2021	Manufacturing digital-twin framework	Framework, not ready-made diagnosis model
Shao & Helu (2020)	Manufacturing digital-twin framework	Does not prove model accuracy in a specific plant
Carvalho et al. (2019)	Review of ML for predictive maintenance	Results depend on data, labels, and context
Dalzochio et al. (2020)	ML and reasoning for predictive maintenance	Deployment and integration context-dependent
NIST ISO 23247 use cases	Implementation scenarios	Do not replace project-specific validation

2.3 Research Gap and Differentiation

Proposed differentiator: connect anomaly to inspectable evidence, show candidate causes with uncertainty, and compare one bounded scenario. This remains a design objective until evaluated.

2.4 Feedback, Evaluation, Suggested Improvements

Lecturer feedback: pending; no assessment supplied. Improvements: expand search after choosing exact use case, add domain-specific sources, compare baseline with stronger model, conduct error analysis, assess uncertainty if supported, gather user feedback if available, revise after instructor review.

2.5 Final Grading Criteria

Guidelines mention documentation, implementation, testing, and presentation but no numeric weights. Request official rubric. Internal checks: completeness, correctness, evaluation, reproducibility, limitations, team contributions, demo quality.

2.6 References

- ISO (2021). ISO 23247-1:2021, Digital twin framework for manufacturing, Part 1. https://www.iso.org/standard/75066.html
- Shao & Helu (2020). Framework for a digital twin in manufacturing. Verify exact bibliographic details.
- Carvalho et al. (2019). A systematic literature review of ML methods applied to predictive maintenance. Computers & Industrial Engineering, 137, 106024. https://doi.org/10.1016/j.cie.2019.106024
- Dalzochio et al. (2020). Machine learning and reasoning for predictive maintenance in Industry 4.0. Computers in Industry, 123, 103298. https://doi.org/10.1016/j.compind.2020.103298
- NIST. Use case scenarios for digital twin implementation based on ISO 23247. Verify exact metadata.

Verify exact metadata for Shao & Helu and NIST entries against original records before submission.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs	Validation
Maintenance engineer	Inspect alerts, evidence, likely causes	Incident walkthrough
Production supervisor	Understand impact, compare safe options	Review scenario assumptions
Quality engineer	Relate anomalies to quality events	Review evidence links
Project team / ML engineer	Reliable pipeline, metrics, reproducibility	Code/metric review
Instructor/evaluator	Scope, justified methods, reproducible demo	Milestone/final review

3.2 User Stories and Use Cases

ID	User story	Acceptance condition
US-01	Select event and inspect signals/records	Event and records shown; missing data flagged
US-02	View model output to prioritize review	Model/version and score/class shown
US-03	See candidate causes linked to evidence	Source IDs shown; hypotheses separated from facts
US-04	Compare bounded scenario	Baseline/scenario/assumptions shown; no machine changes
US-05	Reproduce pipeline from README	Another member runs demo

3.3 Functional Requirements

ID	Requirement	Priority
FR-01	Ingest authorized dataset/simulator export	Must
FR-02	Validate fields, types, ranges, missing values	Must
FR-03	Record source and preprocessing	Must
FR-04	Split train/validation/test appropriately	Must
FR-05	Baseline anomaly detection/classification per labels	Must
FR-06	Report metrics and confusion matrix when possible	Must
FR-07	Select event for investigation	Must
FR-08	Display supporting signals/records	Must
FR-09	Label explanations as hypotheses	Must
FR-10	Show uncertainty when supported; disclose limitations	Should
FR-11	Provide bounded scenario comparison	Should
FR-12	Record model/data versions and evaluation	Should
FR-13	Show missing-data/processing errors	Must
FR-14	Provide setup/execution README	Must
FR-15	Never send commands to production equipment	Must
FR-16	Reviewer notes if in scope	Could

3.4 Non-functional Requirements

ID	Quality	Requirement
NFR-01	Performance	Measure render/process time; set target after profiling
NFR-02	Reliability	Repeatable runs where method permits
NFR-03	Security	Authorized data; no secrets in repository
NFR-04	Privacy	No confidential data without authorization/protection
NFR-05	Usability	Reviewer can select event and inspect evidence

NFR-06	Traceability	Evidence source IDs and model/data versions recorded
NFR-07	Maintainability	Separate data, model, evaluation, UI modules
NFR-08	Portability	Dependencies and tested setup instructions
NFR-09	Safety	Read-only; no autonomous control
NFR-10	Transparency	Document assumptions, limitations, failure cases

3.5 Data Requirements and Constraints

- Potential fields: timestamps, equipment/process IDs, sensor values, labels if available, maintenance logs, quality records.
- Record source, license, units, missingness, label meaning, and limitations.
- Prevent leakage from future/post-failure information unavailable at prediction time.
- Do not assume live factory connectivity or proprietary APIs without permission.

3.6 Acceptance and Change Control

Review requirements at milestones; accept only with recorded test/demo evidence. Track scope/data/model changes in GitHub issues or project tracker and confirm material changes with instructor.